



Better Choice Brands LLC
www.betterchoicebrands.com

SAFETY DATA SHEET

This Safety Data Sheet (SDS) complies with the requirements of the U.S. Federal Occupational Safety and Health Administration Hazard Communication 29 CFR 1910.1200 (g)(2).

Better Choice Brands Clethodim 2EC

1. IDENTIFICATION OF THE PRODUCT & COMPANY

Product Name: Better Choice Brands Clethodim 2EC

Other means of identification: EPA Reg. No. 103087-4-104199

Intended Use: Herbicide

Company Identification:

Better Choice Brands LLC

4252 E Log Providence Road

Columbia, MO 65201

Emergency Information: In the event of a leak, fire or medical emergency involving a human or animal please call INFOTRAC at 1-800-535-5053 or Outside the USA and Canada at 1-352-323-3500, 24/7/365.

2. HAZARD IDENTIFICATION

GHS-US classification

Health and Environmental Hazards:

Eye irritation	Category 2B
Flammable liquids	Category 4
Aspiration Hazard	Category 1

Required Label Elements

Pictograms:



Signal Word: Danger

Hazard Statements:

Combustible liquid.

May be fatal if swallowed and enters airways.

Precautionary Statements:

Prevention:

Wash thoroughly after handling.

Do not eat, drink or smoke when using this product.

Contaminated work clothing should not be allowed out of the workplace.

Avoid release to the environment.

Response:

If swallowed: get medical help.

If on skin: get medical help.

If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

If eye irritation persists: Get medical advice/attention.

In case of fire: Use dry chemical, carbon dioxide or foam to extinguish.

Storage:

Store in a well-ventilated place. Keep cool.

Store locked up

Dispose of in accordance with local/regional/national/international regulations.

3. **COMPOSITION/INFORMATION ON INGREDIENTS**

Component	CAS Number	CONTENT %	Active Ingredient
Clethodim	99129-21-2	24.3-25.9%	X
Solvent naphtha	64742-94-5	<70%	
Other ingredient	PROPRIETARY	Balance	

The exact percentage (concentration) of composition has been withheld as a trade secret.

4. **FIRST AID MEASURES**

Eyes: If in eyes, flush eyes with water for at least 15 minutes. Seek medical attention if irritation persists.

Skin: If skin becomes irritated, wash with soap and water. Seek medical attention if irritation persists.

Ingestion: Quickly rinse mouth out with water. Do not induce vomiting. Seek medical attention or call a poison control center for treatment advice.

Inhalation: If irritation or respiratory symptoms develop, move immediately to fresh air. Seek medical attention if irritation is severe or persists.

5. **FIRE FIGHTING MEASURES**

Extinguishing Media: Water fog, foam, dry chemical or carbon dioxide.

Special Hazards Arising from the Substance/Mixture: Normal combustion forms carbon dioxide, water vapor and may produce oxides of nitrogen, sulfur. Combustion may produce toxic compounds of chlorine.

Special protective equipment and precautions for fire-fighters: A self-contained breathing apparatus and suitable protective clothing must be worn in fire conditions.

Advice for Fire Fighters: Products of combustion from fires involving this material may be toxic. Avoid breathing smoke and mists. Avoid personnel and equipment contact with fallout and runoff. Minimize the amount of water used for firefighting. Do not enter any enclosed area without full protective equipment, including self-contained breathing equipment. Contain and isolate runoff and debris for proper disposal. Decontaminate personal protective equipment and firefighting equipment before reuse. Read the entire document

Additional Information: None.

6. **ACCIDENTAL RELEASE MEASURES**

Personal Precautions: None typically required. Use personal protective equipment as necessary. Follow label directions.

Environmental Precautions: Do not discharge into aquatic environment.

Methods for Containment/Clean-Up: Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Collect and transfer the product into a properly labelled and tightly closed container. Clean contaminated floors and objects thoroughly, observing environmental regulations. Contaminated soil may have to be removed and disposed.

7. **HANDLING AND STORAGE**

Handling: Use only in area provided with appropriate exhaust ventilation. Avoid contact with skin, eyes and clothing. Keep working clothes separately. Wash hands immediately after work, if necessary, take a shower. Remove soiled clothing immediately and clean thoroughly before using again. Garments that cannot be cleaned must be destroyed (burnt). Keep away from: sparks, open flame and direct sunlight.

Storage: Store in the original, unopened container in a cool, dry place, out of direct sunlight and away from stockfeed or foodstuffs. Store in a place accessible by authorized persons only.

8. **EXPOSURE CONTROL/PERSONAL PROTECTION**

Exposure Controls: Follow manufacturer's instructions for cleaning/maintaining PPE. Use in ventilated areas adequate to keep exposure below the TWA. Keep containers closed when not in use.

Personal Protection:

Eye protection: Safety glasses with side-shields.

Skin and body protection: Wear long-sleeved shirt and long pants and shoes plus socks.

Respiratory protection: A respirator is not normally required when handling this substance. Use effective engineering controls to comply with occupational exposure limits. When respirators are required, select a certificated equipment based on actual or potential airborne concentrations and in accordance with the appropriate regulatory standards and/or industry recommendations.

General protective measures: Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables, use detergent and warm/tepid water. Keep and wash PPE separately from other laundry.

9. **PHYSICAL AND CHEMICAL PROPERTIES**

Appearance:	Light yellow to light brown liquid
Odor:	Characteristic aromatic compounds odor at 20.5°C
Melting/Freezing Point:	Not determined
pH:	4.0-7.0
Solubilities:	Not determined
Flash point:	≥60
Density:	0.9714±0.02g/ml
Water content %:	≤0.8
Emulsion stability (20x dilution) 2 h:	Separation of cream or oil observed ≤0.3ml
Persistent foaming (1min), mL:	≤60

10. **STABILITY AND REACTIVITY**

Reactivity: No specific test data related to reactivity available for this product or its ingredients.

Chemical Stability: This product is stable material under normal conditions of storage and handling.

Avoid contact with ignition sources.

Possibility of hazardous reactions: Will not occur hazardous polymerization.

Conditions to avoid: High temperatures and excess moisture. Do not store for prolonged periods in direct sunlight.

Incompatible Materials: Avoid strong oxidizing agents.

Hazardous decomposition products: No decomposition products expected under normal conditions of use.

11. **TOXICOLOGICAL INFORMATION**

Acute oral toxicity rat:	LD ₅₀ =5000 mg/kg b.w
Acute dermal toxicity rabbit:	LD ₅₀ >5000 mg/kg
Acute inhalation toxicity rat(4h):	LC ₅₀ > 5.05 mg/L
Skin irritation rabbit:	Non irritating
Eye irritant rabbit:	Mildly-irritation.
Sensitisation guinea pig:	Non-sensitizing

12. **ECOLOGICAL INFORMATION**

Birds (Coturnix japonica) :	Acute oral LD ₅₀ : > 2000 mg/kg-bwt
Fish (Poecilia reticulata(Guppy)) :	96-hour LC ₅₀ : > 100mg/L
Daphnia:	48-hour LC ₅₀ : > 100 mg/L
Water algae:	72-hour EyC ₅₀ : > 100mg/L
	72-hour ErC ₅₀ : > 100mg/
Honeybees:	Contact LC ₅₀ (48h) :> 100 µg/bee
	Oral LC ₅₀ (48h): >100 µg/bee
Earthworms:	LC ₅₀ (14d): >1000 mg/kg.

13. **DISPOSAL CONSIDERATIONS**

Pesticide disposal: Open dumping or burning of this pesticide is prohibited. Waste resulting from the use of this product that cannot be reused or reprocessed. Never pour untreated waste into public sewers or where there is any danger of run-off or seepage into water systems. Do not contaminate rivers, dams or any other water sources with the product or used containers. Comply with local legislation applying to waste disposal.

Container disposal: Do not re-use empty containers. Triple rinse containers. Consult state and local regulations regarding the proper disposal of container. Follow advice on product label and/or leaflet. TRIPLE RINSE empty containers in the following manner: Invert the empty container over the spray or mixing tank and allow to drain for at least 30 seconds after the flow has slowed down to a drip. Thereafter rinse the container three times with a volume of water equal to a minimum of 10 % of that of the container. Add the rinsing to the contents of the spray tank before destroying the container in the prescribed manner. Do not re-use the empty container for any other purpose but destroy it by perforation and flattening and bury in an approved dumpsite. Prevent contamination of food, feedstuffs, drinking water and eating utensils. Comply with local legislation applying to waste disposal.

14. **TRANSPORTATION INFORMATION**

DOT :	Not Regulated when shipped domestically in individual containers less than 119 gallons
UN/ID no	For packages greater than 119 Gallons: NA1993
Proper shipping name	COMBUSTIBLE LIQUID, N.O.S.
Hazard class	Class 3
Packing group	III
IATA	Not regulated
IMDG	Not regulated

15. **REGULATORY INFORMATION**

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. Following is the hazard information as required on the pesticide label:

signal word DANGER

Harmful if swallowed. Causes eye irritation.

International Inventories

USINV	Not present
DSL/NDL	Not Present
EINECS/	Not Present
ELINCS	Not Present
ENCS	Not Present
China	Not Present
KECL	Not Present
PICCS	Not Present
AICS	Not Present
TSCA	Not Present

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japan Existing and New Chemical Substances
IECSC - China Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
AICS - Australian Inventory of Chemical Substances

SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains no chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

16. OTHER INFORMATION

This MSDS summarizes our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this MSDS and consider the information in the context of the how the product will be handled and used in the workplace including in conjunction with other products.

If clarification or further information is needed to ensure that an appropriate risk assessment can be made the user should contact the company.